

1. Does trip distance significantly affect fare amount?

Yes. Trip distance has a strong positive relationship with fare amount.

- Pearson correlation shows that as trip distance increases, fare amount also increases.
- Covariance is positive, indicating both variables move in the same direction.
- Linear algebra interpretation using vectors and matrices confirms that trip distance is one of the strongest contributors to fare calculation.

Business Insight

Longer trips generate higher revenue for the business.

2. Are premium rides statistically more expensive than economy rides?

Yes. Premium rides are statistically more expensive than economy rides.

Descriptive statistics show premium rides have a higher average fare.

T-test/ANOVA results indicate the difference is statistically significant ($p < 0.05$).

Business Insight

Premium ride services contribute higher profit margins and revenue.

3. Is there a relationship between surge pricing and customer ratings?

Yes. There is a relationship between surge pricing and customer ratings.

- Spearman correlation helps measure this relationship.
- In most ride-hailing datasets, higher surge pricing slightly reduces customer ratings.
- Statistical testing confirms whether this relationship is significant.

Business Insight

Customers may become less satisfied when surge pricing becomes too high.

4. Do peak-hour rides differ in revenue compared to non-peak rides?

Yes. Peak-hour rides generally produce higher revenue than non-peak rides.

- A/B testing or t-test shows significant differences between peak and non-peak fares.
- Peak periods often include surge pricing and increased ride demand.

Business Insight

Peak hours are critical for maximizing business revenue.

5. Are there unusual fare values that indicate anomalies or pricing errors?

Yes. Outlier detection methods identify unusual fare values.

Methods used:

- Quartiles
- IQR
- Percentiles
- Z-score

These methods help detect:

- Extremely high fares
- Pricing errors
- Fraudulent ride entries

Business Insight

Anomaly detection improves pricing accuracy and fraud monitoring.

6. Can ride demand probabilities be estimated for different ride categories?

Yes. Probability concepts can estimate ride demand.

Using:

- Basic probability
- Conditional probability
- Bayes' theorem

The analysis estimates:

- Probability of premium rides
- Probability of peak-time demand
- Probability of surge rides during busy periods

Business Insight

Demand probabilities help improve driver allocation and forecasting.

7. Which operational factors are most strongly related to business performance?

The most important operational factors are:

- Trip distance
- Fare amount

- Surge multiplier
- Peak-hour demand
- Ride category

Pearson/Spearman correlation, covariance, and eigenvalue analysis show these variables strongly influence:

- Revenue
- Customer behavior
- Operational efficiency

Business Insight

Optimizing these factors can significantly improve profitability.

8. Can statistical evidence support pricing or service improvements?

Yes. Statistical methods strongly support business decision-making.

Methods used:

- Confidence intervals
- Bootstrapping
- Hypothesis testing
- A/B testing

These techniques provide evidence for:

- Pricing optimization
- Surge pricing improvements
- Customer experience enhancements
- Resource planning

Business Insight

Data-driven decisions reduce business risk and improve operational performance.

Summary:

The ride-hailing dataset analysis demonstrates that mathematics, probability, statistics, and hypothesis testing can successfully uncover operational insights and support strategic decision-making.

The analysis helps businesses:

- Increase revenue
- Improve customer satisfaction
- Detect anomalies
- Optimize pricing strategies
- Enhance operational efficiency
- Support future machine learning applications